Solve with us 2.2 - Not Graded

This assignment will not be graded and is only for practice.
Key Points: 1

A matrix is in row echelon form if :

• The first non-zero element (the leading entry) in a row is 1.
• The column containing the leading 1 of a row is to the right of the column containing the leading 1 of the row above it.
• Any non-zero rows are always above rows with all zeros.

A matrix is in reduced row echelon form if :

• The first non-zero element in the first row (the leading entry) is the number 1.
• All subsequent non-zero rows must also have their leading entries (i.e. first non-zero entries) as 1 and they should appear to the right of the leading entry in the previous row.
• The leading entry in each row must be the only non-zero number in its column.
• Any non-zero rows are always above rows with all zeros.

Note: Any matrix which is in reduced row echelon form is also in row echelon form.

1 point

Choose the correct options for the matrix $A = \begin{bmatrix}1&2&0\\0&0&1\\0&0&0\end{bmatrix}$

[Hint: Find out the leading entries for each rows and observe whether the matrix is satisfying the conditions of row echelon form and reduced row echelon form.]

○ A is in row echelon form but not in reduced row echelon form.

○ A is in both row echelon form and reduced row echelon form.

○ A is neither in row echelon form nor reduced row echelon form.

No, the answer is incorrect.
Score: 0
Accepted Answers:
A is in both row echelon form and reduced row echelon form.

1 point

Choose the correct options for the matrix $A = \begin{bmatrix}0&1&0\\1&0&1\\0&1&0\end{bmatrix}$

○ A is in row echelon form but not in reduced row echelon form.

○ A is in both row echelon form and reduced row echelon form.

○ A is neither in row echelon form nor reduced row echelon form.

No, the answer is incorrect.
Score: 0
Accepted Answers:
A is neither in row echelon form nor reduced row echelon form.

1 point

Choose the correct options for the matrix $F = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

- [ ] A is in row echelon form but not in reduced row echelon form.
- [ ] A is in both row echelon form and reduced row echelon form.
- [ ] A is neither in row echelon form nor reduced row echelon form.

No, the answer is incorrect.
Score: 0
Accepted Answers:
A is in row echelon form but not in reduced row echelon form.

Key Points: 2

Let $Ax = b$ be a system of linear equations and suppose A is in reduced row echelon form. Assume that for every zero row of A, the corresponding entry of b is also 0 (i.e., if the $i-th$ row of A is 0, then so is b_i.)

• If the $i-th$ column has the leading entry of some row, we call x_i a dependent variable.
• If the $i-th$ column does not have the leading entry of any row, we call x_i an independent variable.

1 point

Consider a system of linear equations:

$$\begin{aligned} 0x_1 + x_2 + 0x_3 + 0x_4 &= 1 \\ 0x_1 + 0x_2 + x_3 + 0x_4 &= 1 \end{aligned}$$

Choose the set of correct options.
[Hint: Recall the definitions of independent and dependent variable with respect to reduced row echelon form.]

- [ ] x_1 and x_2 are dependent variables.
- [ ] x_2 is a dependent variable.
- [ ] x_3 and x_4 are independent variables.
- [ ] x_4 is an independent variable.

No, the answer is incorrect.
Score: 0
Accepted Answers:
x_2 is a dependent variable.
x_4 is an independent variable.

1 point

let $Ax = b$ be a system of linear equations, where $A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$, and

$b = \begin{bmatrix} 3 \\ 2 \\ 0 \\ 0 \end{bmatrix}$.

Choose the set of correct options.

- [ ] x_1 is a dependent variable.
- [ ] x_2 is a dependent variable.
- [ ] x_3 is a dependent variable.
- [ ] The expressions $x_1 = 3$ and $x_3 = 2 - x_4$ show the dependency of x_1 and x_3 on the elements of b and the independent variables.

No, the answer is incorrect.
Score: 0
Accepted Answers:

x_1 is a dependent variable.
x_3 is a dependent variable.
The expressions $x_1 = 3$ and $x_3 = 2 - x_4$ show the dependency of x_1 and x_3 on the elements of b and the independent variables.

Key Points: 3

The three different types of elementary row operations that can be performed on a matrix are:

- **Type 1:** Interchanging two rows.
- **Type 2:** Multiplying a row with some constant.
- **Type 3:** Adding a scalar multiple of a row to another row.

Using these row operations one can obtain row echelon form or reduced row echelon form of a given matrix.

Consider the following matrix A and answer the questions (a), (b), (c), (d), and (e).

$$\begin{bmatrix} 1 & 3 & 2 & 0 \\ 0 & 2 & 1 & 0 \\ 2 & 4 & 4 & 3 \end{bmatrix}$$

1 point

What will be the third row of the matrix if we perform the row operation $R_3 - 2R_1$ on A?

- ( ) (1123)

○ (41083)(41083)

○ (0−203)(0−203)

○ (0443)(0443)

No, the answer is incorrect.
Score: 0
Accepted Answers:
(0−203)(0−203)

1 point

What will be the second row of the matrix if we perform the row operation $\frac{1}{2}R_2$ 21R2 on AA?

○ (0110)(0110)

○ (0420)(0420)

○ $(0 1 \frac{1}{2} 0)$(01210)

○ $(1 2 2 \frac{3}{2})$(12223)

No, the answer is incorrect.
Score: 0
Accepted Answers:
$(0 1 \frac{1}{2} 0)$(01210)

1 point

If we perform the row operations mentioned in question (a) and question (b), simultaneously on AA, the matrix (let it be called BB) we get is:

○ $\begin{bmatrix} 1 & 3 & 2 & 0 \\ 0 & 1 & \frac{1}{2} & 0 \\ 0 & -2 & 0 & 3 \end{bmatrix}$ 10031−22210003

○ $\begin{bmatrix} 1 & 3 & 2 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 2 & 3 \end{bmatrix}$ 101311212003

○ $\begin{bmatrix} 1 & 3 & 2 & 0 \\ 0 & 4 & 2 & 0 \\ 0 & -2 & 0 & 3 \end{bmatrix}$ 10034−2220003

○ $\begin{bmatrix} 1 & 3 & 2 & 0 \\ 0 & 1 & \frac{1}{2} & 0 \\ 0 & 4 & 4 & 3 \end{bmatrix}$ 1003142214003

No, the answer is incorrect.
Score: 0
Accepted Answers:

$$\begin{bmatrix} 1 & 3 & 2 & 0 \\ 0 & 1 & \frac{1}{2} & 0 \\ 0 & -2 & 0 & 3 \end{bmatrix}$$

1 point

Which row operations have to be performed on the matrix B (matrix B is as in question (c)) to make all the entries 0 in the second column, except the leading entry of the second row.

- ◯ $R_1 - 3R_2$ and $R_3 - 2R_2$
- ◯ $R_1 + 3R_2$ and $R_3 + 2R_2$
- ◯ $R_1 + 3R_2$ and $R_3 - 2R_2$
- ◯ $R_1 - 3R_2$ and $R_3 + 2R_2$

No, the answer is incorrect.
Score: 0
Accepted Answers:

$R_1 - 3R_2$ and $R_3 + 2R_2$

1 point

The reduced row echelon form of A is

[Hint: Recall the definition of reduced row echelon form.]

◯

$$\begin{bmatrix} 1 & 0 & \frac{1}{2} & 0 \\ 0 & 1 & \frac{1}{2} & 0 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

◯

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

◯

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -\frac{3}{2} \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

◯

$$\begin{bmatrix} 1 & 0 & 0 & -\frac{3}{2} \\ 0 & 1 & 0 & -\frac{3}{2} \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

No, the answer is incorrect.
Score: 0
Accepted Answers:

$$\begin{bmatrix} 1 & 0 & 0 & -\frac{3}{2} \\ 0 & 1 & 0 & -\frac{3}{2} \\ 0 & 0 & 1 & 3 \end{bmatrix}$$ 100010001−23−233

Key Points: 4

Let $Ax = b$Ax = b denote the matrix representation of a system of linear equations.

- The augmented matrix is denoted by $[A|b]$[A|b].
- The matrix obtained by performing the operations that transform AA to its reduced row echelon form RR on the augmented matrix is denoted by $[R|c]$[R|c].
- The solutions of $Rx = c$Rx = c are the same as the solutions of $Ax = b$Ax = b.

Consider a system of linear equations

$2x_1 + x_2 = 1 - x_1 + x_3 + x_4 = -1x_1 + x_2 - x_3 + x_4 = 2 - x_1 + x_3 + x_4 = 1.$2x1 + x2 = 1−x1 + x3 + x4 = −1x1 + x2 − x3 + x4 = 2−x1 + x3 + x4 = 1.

Answer the questions (a), (b) and (c) based on the above information.

1 point
Which of the following represents an augmented matrix of the system?

[Hint: First try to write down the coefficient matrix AA and the corresponding vector bb.]

○
○
○
○

No, the answer is incorrect.
Score: 0
Accepted Answers:

1 point
What will be the matrix $[R|c]$[R|c] obtained by performing the operations that transform AA to its reduced row echelon form RR on the augmented matrix?

○
○
○
○

No, the answer is incorrect.
Score: 0
Accepted Answers:

1 point
Choose the correct option. [Hint: Write down the system of linear equations corresponding to the reduced row echelon form of the augmented matrix.]

○ The system of linear equations has an infinite number of solutions.
○ The system of linear equations has a unique solution.
○ The system of linear equations has no solution.
○
$x_1 = x_2 = x_3 = x_4 = 1$x1 = x2 = x3 = x4 = 1 is a solution of the system of linear equations.

No, the answer is incorrect.
Score: 0
Accepted Answers:
The system of linear equations has no solution.

Check Answers